

Time Series Analysis: S&P 500 Stock Market Closing Prices

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Author Note

Thank you to all of the support I have received from my professor, TA, and my classmate in this course. No matter what grade result, I will always learn from these experiences to develop my work ethic and approach. For this project, I chose to perform a time series analysis on the S&P 500 Index. I have always wanted to try working with stock market predictions, so I chose to give this an attempt. During this time, I was struggling with my own health, but I tried to do everything to my best capability. After completing this project, I have learned a lot through my work, and I plan to grow my skills in the future.

The authors made the following contributions. Joseph Shin: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing, Coding, Interpretations.

Abstract

15

16 In this study, the main objective is to forecast the closing prices of the S&P 500 stock
17 index using Seasonal Autoregressive Integrated Moving Average, otherwise known as
18 SARIMA. Since stock market prices have a complex pattern influenced by economic
19 factors, investor behavior, market trends, and real-life events, I want to accurately forecast
20 the future closing prices by performing model transformations and model diagnostics to get
21 a more accurate result.

22 As a result, the next five S&P 500 closing price predicted values after October 29th,
23 2024, are \$5,832.61, \$5,832.13, \$5,835.73, \$5,835.73, and \$5,835.73, and the standard error
24 of these values are \$30.48, \$41.59, \$50.67, \$58.26, \$65.00. This result reveals that there is a
25 stable pattern around \$5,832 - \$5,835 for the S&P 500 stock prices, while the standard
26 errors slowly increases in the future. The fact that the next five forecasted values seems
27 stable rather than showing unexpected fluctuations or continued trends is a surprise.

28 Due to the stability, this may imply that the market is much more resilient and the
29 investors are more consistent and confident during this forecasting period. What could this
30 mean? This could mean that economic conditions are more steady and the top 500 leading
31 companies are performing at a consistent pace. Additionally, since the forecast is stable, it
32 may tell investors to not change their current investment strategy. The increasing
33 uncertainty over time makes sense as time goes on, since the natural variability may
34 increase as well. The increasing variance exemplifies the significance of unpredictability.

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Introduction

The problem that I am issuing from this S&P 500 data set is to determine what the future values of the market's closing price will be after October 29th, 2024. The data provides the daily index value at market close, and the market closing time is at 4 pm ET. The index includes the top 500 leading companies in the leading industries of the U.S. economy, and this covers about 75% of the U.S. equities. The data shows the market's overall health and trends, which is important for investors, economists, and policymakers in how they make their decisions based on the pattern. The original data set used to have missing dates and values, but I put those missing dates back in and made the values equal to the previous date's closing price value. I also had to change the S&P 500 observations for the dates that had no information and fill those in with the previous date's closing price value as well.

<https://fred.stlouisfed.org/series/SP500#>

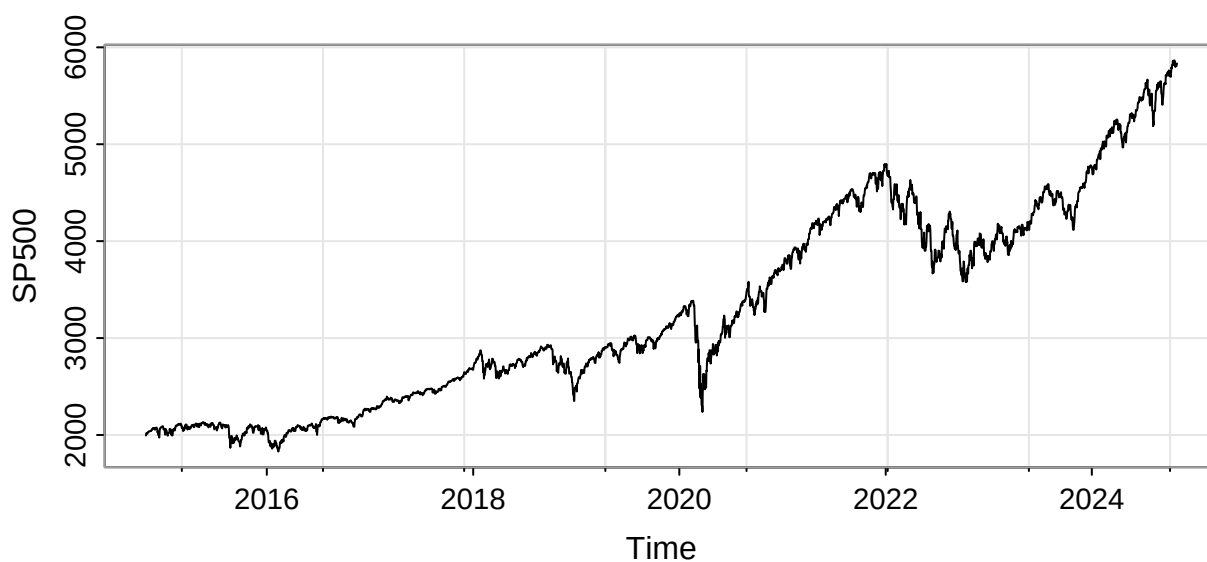
(S&P Dow Jones Indices LLC)

Methods

```
## Warning: package 'forecast' was built under R version 4.3.3
```

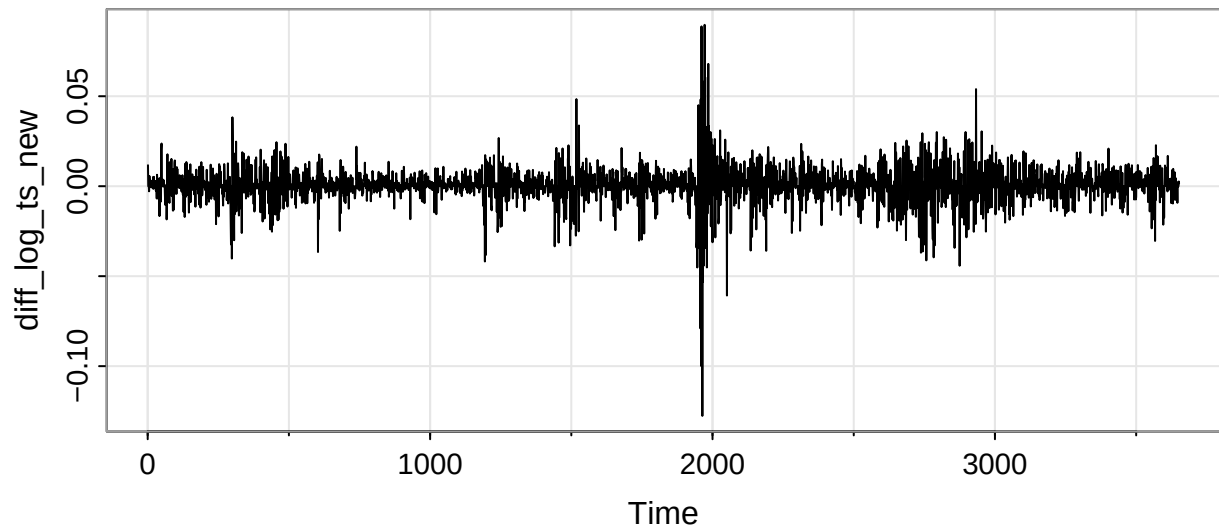
```
## Warning: package 'tseries' was built under R version 4.3.3
```

For my packages, I utilized forecast, tseries, astsa, readr, dplyr, lubridate, and tidy. To start off, I had to clean the data set first. First, I changed the dates into a date format and filled in the missing dates. Then, for the dates with missing S&P 500 closing price value, I set those values to the previous day's closing price value.



```
##  
## Augmented Dickey-Fuller Test  
##  
## data: SP500  
## Dickey-Fuller = -2.2249, Lag order = 15, p-value = 0.4831  
## alternative hypothesis: stationary
```

Next, I had to make sure my time series was stationary. As you can see in the graph above, it is not stationary. I even tried using the `adf.test()` function to test the stationarity, and it rejected the null hypothesis since $0.4831 > 0.05$ (H_0 = Time series is stationary). Therefore, I had to transform the model. The transformation I chose to perform was log-transformation. Additionally, I took the difference of the log transformed model. Here are the results:



71

```
72 ## Warning in adf.test(diff_log_ts_new): p-value smaller than printed p-value
```

```
73 ##
```

```
74 ## Augmented Dickey-Fuller Test
```

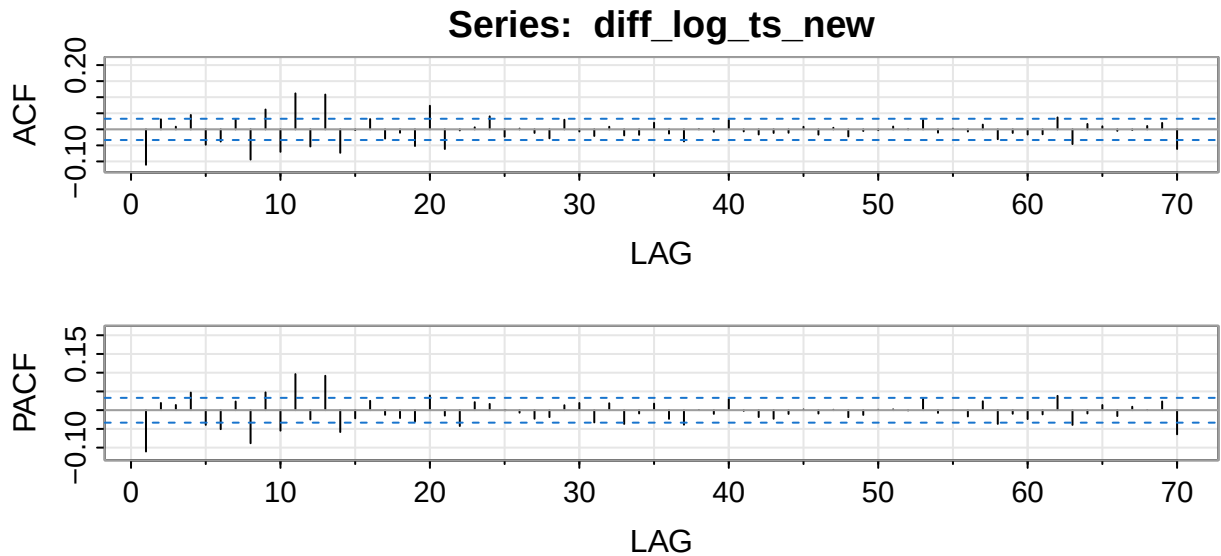
```
75 ##
```

```
76 ## data: diff_log_ts_new
```

```
77 ## Dickey-Fuller = -14.827, Lag order = 15, p-value = 0.01
```

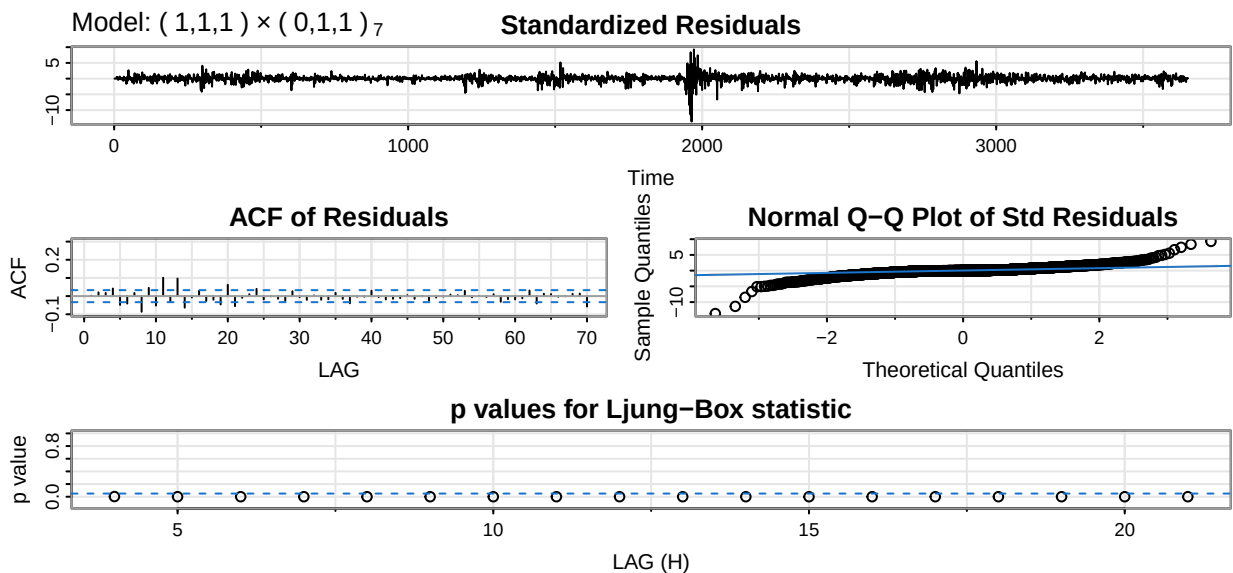
```
78 ## alternative hypothesis: stationary
```

79 As you can see in the graph, white noise is present. Additionally, the mean stays
80 constant around a smaller range of values, and the variance does not fluctuate. The
81 `adf.test()` function also proves that it is stationary since the p-value is smaller than 0.01.
82 Therefore, I chose to continue with this differenced log-transformed model since it was
83 stationary.

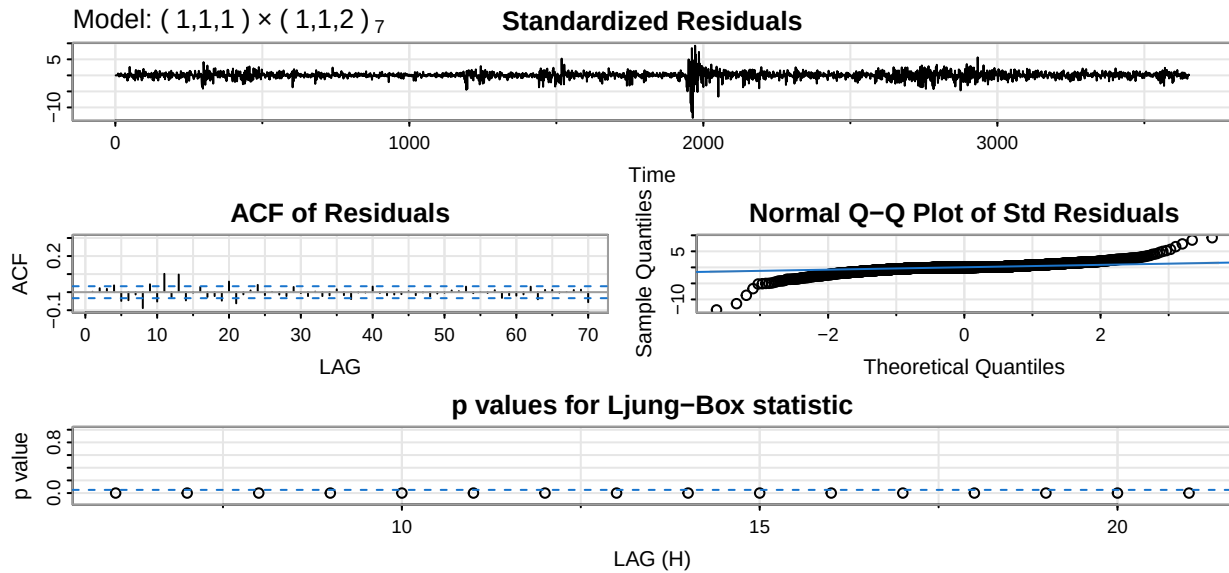


84

85 As you can see in the plots above, the ACF and PACF are both tailing off. This
 86 suggests the presence of an ARMA process. Therefore, I tested out multiple different
 87 combinations of p , d , q , and seasonal P , D , Q for my `sarima` function to determine which
 88 combination gave the best result. After testing multiple different models, I narrowed it
 89 down to two different models. One of them was where $p=1$, $d=1$, $q=1$, $P=0$, $D=1$, and
 90 $Q=1$. The other model consisted of $p=1$, $d=1$, $q=1$, $P=1$, $D=1$, and $Q=2$. For both
 91 models, I had to set seasonality equal to 7. I was originally going to set seasonality to 365,
 92 but my code kept crashing, so I chose to go for a weekly seasonality using the daily data
 93 set instead.

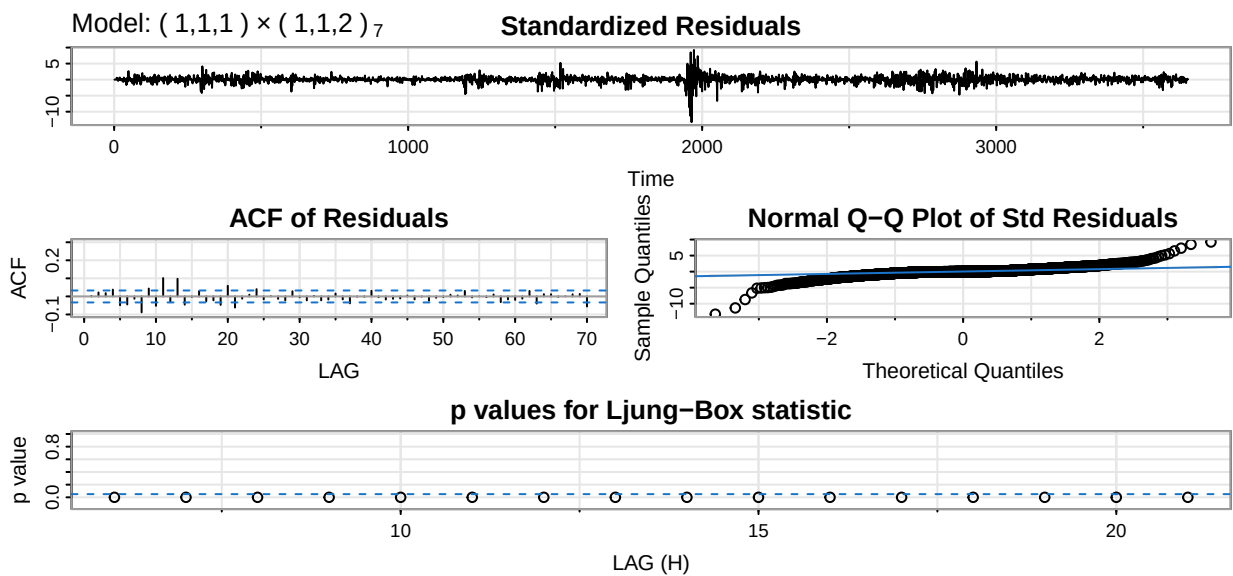


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95

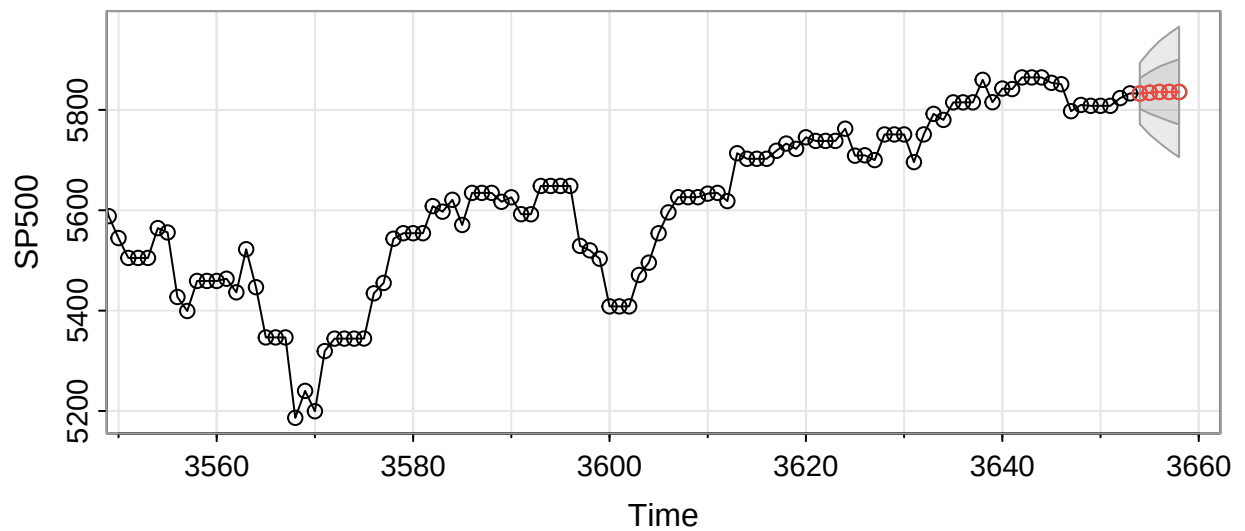
96 The reason that these two models were the best model suggestions was because they
 97 had the lowest AIC compared to all of the other models I tested. Looking at the summary
 98 statistics of these models, they have almost identical results. Because of that, I chose to
 99 just commit with one final model, which was snp_2, since it had a lower AIC compared to
 100 snp_1.



101

102 Based on the standardized residuals plot, there is white noise present, and the
 103 residuals are generally centered around zero, which hints that the model is capturing the
 104 overall trend.

Results



```

107 ## $pred
108 ## Time Series:
109 ## Start = 3654
110 ## End = 3658
111 ## Frequency = 1
112 ## [1] 5832.613 5834.128 5835.730 5835.735 5835.734
113 ##
114 ## $se
115 ## Time Series:
116 ## Start = 3654
117 ## End = 3658
118 ## Frequency = 1
119 ## [1] 30.47526 41.59328 50.67363 58.26477 64.99951

```

120 When performing my forecast, I used the same p, d, q, and seasonal P, D, Q values,
 121 but switched to the original dataset because I wanted it to print out the correctly scaled
 122 S&P 500 closing prices. The model's performance captures the trends, especially through

short-term forecasts. The confidence intervals give a good idea of what the range of uncertainty is, which is important for investor in how they make their decisions. It is important for them to not change their methods in how they invest because the values seem to be stabilized during this period. However, the increasing uncertainty confidence interval is something that they should be cautious of since there may be unexpected drops in the stock market.

Discussion

Overall, the next five predicted values of the S&P 500 daily index of the closing prices are stable. One limitation that my study suffered from was that I had to perform the model with seasonality equal to 7 instead of 365. I think setting the seasonality to 365 would have provided more accurate results. While the model performance went fine, it could benefit from the lack of these limitations.

References

135

- 136 S&P Dow Jones Indices LLC, S&P 500 [SP500], retrieved from FRED, Federal
137 Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/SP500>, December 20, 2024.